

Shivani Bakshi

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EXPERIENCE

ThoughtSpot

Delhi, India

Senior Software Engineer | Member of Technical Staff 4

Nov. 2022 – Present

- Engineered Muze, a browser-based JavaScript visualization framework powering exploratory analytics and interactive data visualizations.
- Designed and implemented Valkyrie, a middleware architecture that decoupled Muze from ThoughtSpot's core platform, enabling pluggable visualization engine integration.
- Architected TypeScript and MobX state management infrastructure coordinating bidirectional communication between application state and visualization rendering pipelines.
- Extended a Rust-based data processing layer compiled to WebAssembly, implementing browser-side data transformations including null handling and 100% stacked chart support.
- Established automated testing infrastructure using Playwright, Cypress, and GitHub Actions, maintaining 90% test coverage across critical workflows.

Pixis.ai

Bangalore, India

Software Engineer

Sep. 2020 – Nov. 2022

- Developed Pixis CoCreate, a no-code advertising platform supporting multi-variant workflows and drag-and-drop creative generation.
- Built scalable React and TypeScript components for campaign management workflows used by marketing teams.
- Extended a Chrome extension integrated with Facebook Ads Manager, enabling bulk campaign operations and optimization workflows.
- Developed frontend systems for digital asset management, improving creative production and collaboration workflows.

ThoughtFocus – Blackstone COE

Gurgaon, India

Software Engineering Intern

Jan. 2020 – Jul. 2020

- Developed a C# .NET automation tool integrating Jira REST APIs to extract and process project tracking data.
- Designed data persistence workflows using Dapper ORM and SQL, replacing manual reporting processes.

PROJECTS

Hybrid Analytical Query Engine | Rust, WebAssembly, TypeScript, React, Node.js

[GitHub](#)

- Built an in-browser analytical query engine using Rust compiled to WebAssembly, enabling privacy-preserving data processing without server round trips.
- Implemented a SQL-like query execution pipeline supporting filtering, aggregations, grouping, and sorting over client-side datasets.
- Developed a React-based analytics interface for dataset exploration, visualization, and query performance benchmarking.

Enterprise Knowledge Assistant (RAG) | Python, FastAPI, React, PostgreSQL, pgvector

In Progress

- Building an enterprise document intelligence platform using Retrieval-Augmented Generation (RAG) to enable semantic search and context-aware answers over internal knowledge bases.
- Designing an end-to-end document processing pipeline covering ingestion, chunking, embedding generation, and vector similarity search using pgvector.
- Implementing FastAPI backend services and a React interface for document upload, search, and conversational interactions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, C++, Python, SQL

Frontend: React.js, Next.js, MobX, Tailwind CSS, HTML5, CSS3

Backend & Data: Node.js, FastAPI, REST APIs, PostgreSQL, pgvector, MySQL

Systems & AI: WebAssembly (Rust), RAG Pipelines, Vector Search, Embeddings

Tools & Cloud: Vercel, Git, GitHub Actions, Playwright, Cypress

EDUCATION

Indira Gandhi Delhi Technical University for Women

Delhi, India

Master of Computer Applications

Aug. 2017 – Jul. 2020

University of Delhi

Delhi, India

Bachelor of Science in Computer Science

Aug. 2014 – Jul. 2017